Name:- Shivendera Pratap Singh
Superset ID:- 6399051
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🧠 MCP Quiz (Model Context Protocol)

◆ Section A: MCQs (1 Mark Each)

1. What is MCP primarily used for?
A. Data visualization
B. Agent-to-tool communication
C. Database management
D. UI design

2. MCP stands for:
A. Model Control Pipeline
B. Machine Context Processing
C. Model Context Protocol
D. Managed Compute Platform

3. In MCP architecture, what is the role of an "Agent"?
A. Stores data
B. Executes database queries
C. Decides actions and interacts with tools
D. Designs UI

4. Which component actually performs tasks like API calls?
A. Agent
B. Tool
C. Context
D. Connector

5. MCP helps in:
A. Replacing APIs

B. Standardizing communication between AI and tools
C. Removing cloud dependency
D. Building databases

◆ Section B: True / False (1 Mark Each)

6. MCP eliminates the need for tools. **False**

7. Agents can call multiple tools using MCP. **True**

8. MCP only works with Python. **False**

9. Context in MCP helps the agent make better decisions. **True**

10. MCP architecture includes Agent, Tool, and Context. **True**

◆ Section C: Short Answer (2 Marks Each)

11. Define Model Context Protocol (MCP).

Answer:-

Model Context Protocol (MCP) is a standardized framework that enables AI agents to interact with external tools and systems efficiently by structuring communication, context sharing, and execution workflows.

12. What is "context" in MCP?

Answer:-

Context is the structured information (user input, history, metadata, environment state) that helps the agent make informed decisions and choose appropriate actions/tools.

13. Give two examples of MCP tools.

Answer:-

- Weather API tool
- Database query tool
  (others: calculator, search API, payment gateway)

14. What is the difference between an Agent and a Tool in MCP?

Answer:-

Agent	Tool
Decision-maker	Executor
Uses reasoning (LLM)	Performs specific task
Chooses which tool to call	Executes API/function

15. Why are connectors important in MCP?

Answer:-

Connectors act as bridges between agents and tools, ensuring:

- Standardized communication
- Compatibility across systems
- Secure and reliable data exchange

◆ Section D: Scenario-Based Questions (3 Marks Each)

16. Scenario:
An AI assistant receives a user query: *"What is the weather in Delhi?"*

- Identify the role of the agent
- Identify the tool used
- Explain how MCP enables this interaction

Answer:-

Agent Role:

- Understands query ("weather in Delhi")
- Decides to call weather tool

Tool Used:

- Weather API

How MCP enables this:

- Structures request → agent selects tool
- Passes parameters (location = Delhi)
- Tool returns result → agent formats response

17. Scenario:
A chatbot needs to fetch customer data from a database and then calculate a discount.

- How many tools are involved?
- What is the role of MCP here?

Answer:-

Number of tools:
- 2 tools
    - Database tool (fetch customer data)
    - Calculation tool (discount logic)

Role of MCP:
- Coordinates multi-step workflow
- Maintains context between steps
- Ensures correct sequencing of tool calls

18. Scenario:
An agent fails to provide accurate output due to missing context.

- What went wrong?
- How can MCP fix this issue?

Answer:-

What went wrong:
- Agent lacked sufficient context (e.g., user history, preferences)
- Result → poor or incorrect decision

How MCP fixes it:
- Maintains structured context
- Provides relevant data to agent before decision
- Enables better reasoning and tool selection

◆ Section E: Advanced / Conceptual (5 Marks Each)

19. Explain MCP architecture with its key components.

Answer:-

Core Components:
1. Agent
    - Brain of the system (LLM-driven)

    - Interprets user input
    - Decides actions
1. Tool
    - Executes tasks (API, DB, computation)
    - Returns structured outputs
1. Context
    - Shared memory/state
    - Includes user query, history, system data
1. Connector (optional but critical)
    - Standard interface layer
    - Handles communication between agent and tools

20. Describe how Agent–Tool communication works in MCP step by step.

- User Input
- User sends query
- Context Creation
- MCP builds structured context (query + history)
- Agent Reasoning
- Agent analyzes context
- Decides which tool is needed
- Tool Selection
- MCP formats request for selected tool
- Tool Execution
- Tool performs task (API call, DB query, etc.)
- Response Return
- Tool sends structured output back
- Context Update
- MCP updates context with results
- Final Response
- Agent interprets result and responds to user